

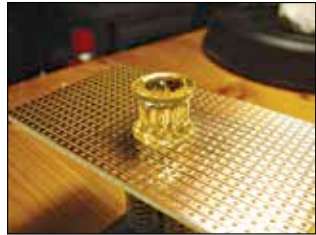
The most important thing to remember is that the 3D object is printed layer by layer. This means that the .stl file needs to be converted in to a lot of individual layers as a G-code file. Here is where Slic3r and Skeinforge come in to play. While Slic3r is newer and isn't as reliable as Skeinforge, it's much easier to use and spits out G-codes faster than you can say Fused Deposition Modelling (only kidding). Also, unlike Slic3r, Skeinforge needs Python to be installed on your computer to run it.

With your G-code file ready, the only step left is to send it to the printer and here's where ReplicatorG comes in to use. This is the same software that is used by MakerBots, RepRaps and CNC machines alike. With tens of thousands of successful prints, this would be your best bet.

Other noteworthy mentions would be Red Snapper and Printron or Pronterface.

Bonus: Other 3D printer types and their plastics

While printers like out RepRap Prusa Mendel and quite a few others use filaments and follow a process called Fused Deposition Modelling (FDM), but there are a few DIY printers out there. Stereolithography is where you use a laser to heat and solidify a resin to build a model



Photoplastic!

Formlabs' Form 1 being a good example and instead of a bottom up approach to printing used by the RepRap, The SLA (Stereolithographic Apparatus) takes a top down approach. The resin is a photoplastic that cures with the help of UV light and the SLA requires a tank full of that stuff.

A small portion on the surface of the bath is cured. A perforated platform slowly lowers itself across the depth of the tank to expose more resin, layer by layer to the laser.

The point in favour of SLAs is the fact that your model will be ready much faster when compared to an FDM printer but it's going to set you back a lot more. The resin alone would set you back about a \$100 for a litre and built machines would cost as much as a Mercedes.

There are a few DIY projects but it tends to get a little complicated and isn't as straightforward as the RepRap.

Other 3D printing technology range from the use of electron beams to carbon dioxide lasers and ytterbium fiber lasers (are you s#!++ing me?) and